

6A.2 Pressure difference required for flow in pipe with elevation change (Fig. 6A.2). Water at 68°F is to be pumped through 95 ft of standard 3-in. pipe (internal diameter 3.068 in.) into an overhead reservoir.

(a) What pressure is required at the outlet of the pump to

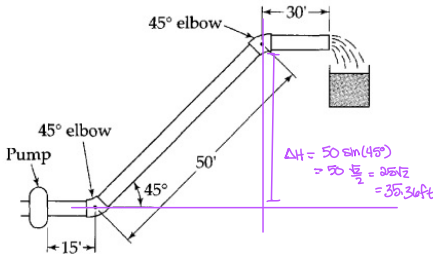


Fig. 6A.2. Pipe flow system.

supply water to the overhead reservoir at a rate of 18 gal/min? At 68°F the viscosity of water is 1.002 cp and the density is 0.9982 g/ml.

(b) What percentage of the pressure drop is needed for overcoming the pipe friction?

Answer: (a) 15.2 psig

$$a) F_k = (P_0 - P_L) \frac{\pi D^2}{4} \quad F_k = \pi D L \cdot f \cdot \frac{1}{2} \rho V^2$$

$$(P_0 - P_L) \frac{\pi D^2}{4} = \pi D L \cdot f \cdot \frac{1}{2} \rho V^2$$

$$(P_0 - P_L) = 2 \frac{L}{D} f \rho V^2$$

$$= 2 \frac{102.68 \text{ ft}}{0.256 \text{ ft}} (1.00625) (62.36 \frac{\text{lb}}{\text{ft}^3}) (0.779 \frac{\text{ft}}{\text{s}})^2$$

$$P_0 - P_L + (h_0 - h_L) \rho g = 189.8 \frac{\text{lbm}}{\text{ft} \cdot \text{s}^2}$$

$$P_0 - P_L = 189.8 \frac{\text{lbm}}{\text{ft} \cdot \text{s}^2} - \rho g (h_0 - h_L)$$

$$= 189.8 \frac{\text{lbm}}{\text{ft} \cdot \text{s}^2} - (62.36 \frac{\text{lb}}{\text{ft}^3}) (32.174 \frac{\text{ft}}{\text{s}^2}) (-35.26 \text{ ft})$$

$$= 189.8 \frac{\text{lbm}}{\text{ft} \cdot \text{s}^2} - (-70945.26 \frac{\text{lbm}}{\text{ft} \cdot \text{s}^2})$$

$$= 71135.066 \frac{\text{lbm}}{\text{ft} \cdot \text{s}^2} \cdot \frac{1 \text{ ft}}{144 \text{ in}^2} \cdot \frac{1}{32.174 \frac{\text{ft}}{\text{s}^2}} \cdot \frac{1}{144 \frac{\text{in}^2}{\text{ft}^2}} \cdot \frac{1}{144 \frac{\text{in}^2}{\text{ft}^2}}$$

$$= 15.35 \frac{\text{lb}_f}{\text{in}^2}$$

$$= 15.35 \text{ psig}$$

$$b) P_{\text{friction}} = 189.8 \frac{\text{lbm}}{\text{ft} \cdot \text{s}^2}$$

$$P_{\text{tot}} = 71135.066 \frac{\text{lbm}}{\text{ft} \cdot \text{s}^2}$$

$$\% = \frac{\Delta P_{\text{friction}}}{\Delta P_{\text{total}}} \times 100 = \frac{189.8}{71135.066} \times 100 = 0.267\%$$

This means that almost the entirety of the pressure difference needed is to carry the water up the incline.

$$Q = 18 \text{ gal/min}$$

$$D = 3.068 \text{ in} = 0.256 \text{ ft}$$

$$L = 95 \text{ ft} + 15 D + 15 D = 95 \text{ ft} + 30(0.256 \text{ ft}) = 102.68$$

$$\mu = 1.002 \text{ centipoise} \cdot \frac{0.000672 \frac{\text{lb}}{\text{ft} \cdot \text{s}}}{1 \text{ centipoise}}$$

$$= 6.733 \times 10^{-4} \frac{\text{lb}}{\text{ft} \cdot \text{s}}$$

$$\rho = 0.9982 \frac{\text{g}}{\text{ml}} \cdot \frac{1 \text{ ml}}{1 \text{ cm}^3} \cdot \frac{1 \text{ cm}^3}{(0.03937 \text{ ft})^3} \cdot \frac{0.00220462 \text{ lb}}{1 \text{ g}}$$

$$= 62.36 \frac{\text{lb}}{\text{ft}^3} \cdot \frac{0.133681 \text{ ft}^3}{1 \text{ gal}} = 8.336 \frac{\text{lb}}{\text{gal}}$$

$$V = \frac{Q}{A} = \frac{18 \frac{\text{gal}}{\text{min}}}{\pi (0.256 \text{ ft})^2} \cdot \frac{1 \text{ min}}{60 \text{ s}} \cdot \frac{0.133681 \text{ ft}^3}{1 \text{ gal}} = 0.779 \frac{\text{ft}}{\text{s}}$$

$$Re = \frac{D \rho V}{\mu} = \frac{(0.256 \text{ ft}) (62.36 \frac{\text{lb}}{\text{ft}^3}) (0.779 \frac{\text{ft}}{\text{s}})}{6.73344 \times 10^{-4} \frac{\text{lb}}{\text{ft} \cdot \text{s}}} = 18,469.13$$

$$f = \frac{0.0729}{Re^{0.25}} = \frac{0.0729}{(18,469.13)^{0.25}} = 6.25 \times 10^{-3}$$

$$= 0.00625$$

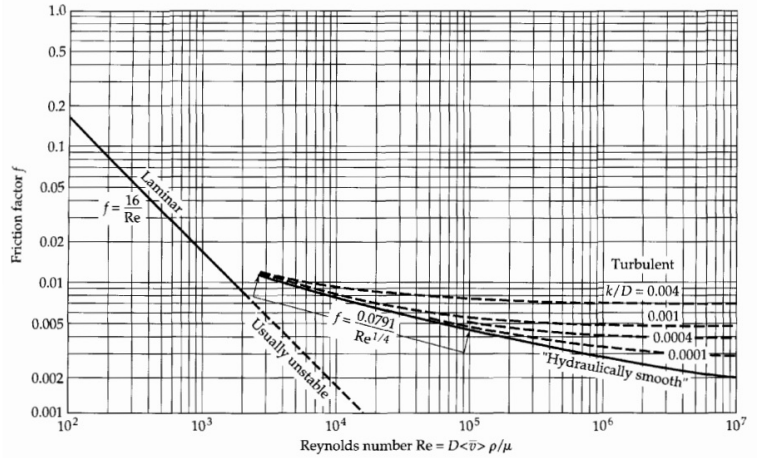


Fig. 6.2-2. Friction factor for tube flow (see definition of f in Eqs. 6.1-2 and 6.1-3. [Curves of L. F. Moody, Trans. ASME, 66, 671-684 (1944) as presented in W. L. McCabe and J. C. Smith, Unit Operations of Chemical Engineering, McGraw-Hill, New York (1954).]

6A.3 Flow rate for a given pressure drop. How many gal/hr of water at 68°F can be delivered through a 1320-ft length of smooth 6.00-in. i.d. pipe under a pressure difference of 0.25 psi? Assume that the pipe is "hydraulically smooth."

(a) Solve by Method A of Example 6.2-2.

(b) Solve by Method B of Example 6.2-2.

Answer: 68 U.S. gal/min

Method A Trial & Error

$$\langle v \rangle = \sqrt{\frac{\Delta P D g}{2 f \rho L}} \quad Re = \frac{\rho D \langle v \rangle}{\mu} \quad f = \frac{0.0791}{Re^{1/4}}$$

From the moody diagram, a smooth pipe between

$\sim 2100 < Re < 10^5$ follows a f_f factor equation:

$$f = \frac{0.0791}{Re^{1/4}}$$

To solve this, excel was used. An initial guess for a friction factor for was made (0.005). $\langle v \rangle$ was calculated.

$\langle v \rangle$ was used to calculate Re . This Re number was used to calculate a new f . This was done until $f_{new} = f$.

F	G	H	I
f	$\langle v \rangle$, m/s	Re	f, new
0.005	0.839185154	3.88E+04	5.64E-03
5.64E-03	0.790483768	3.66E+04	5.72E-03
5.72E-03	0.78459828	3.63E+04	5.73E-03
5.73E-03	0.783865682	3.63E+04	5.73E-03
5.73E-03	0.783774155	3.63E+04	5.73E-03
5.73E-03	0.783762715	3.63E+04	5.73E-03
5.73E-03	0.783761285	3.63E+04	5.73E-03
5.73E-03	0.783761107	3.63E+04	5.73E-03
5.73E-03	0.783761084	3.63E+04	5.73E-03
5.73E-03	0.783761082	3.63E+04	5.73E-03
5.73E-03	0.783761081	3.63E+04	5.73E-03
5.73E-03	0.783761081	3.63E+04	5.73E-03
5.73E-03	0.783761081	3.63E+04	5.73E-03
5.73E-03	0.783761081	3.63E+04	5.73E-03

	A	B	C	D	E	F	G	H	I
1	Parameters					f	$\langle v \rangle$, m/s	Re	f, new
2						0.005	=SQRT(\$C\$5*\$C\$6*\$C\$9/2/\$C\$7/F2/\$C\$3)	= \$C\$6*\$C\$7*G2/\$C\$8	=0.0791/H2^0.25
3	L, ft, m		1320			=12	=SQRT(\$C\$5*\$C\$6*\$C\$9/2/\$C\$7/F3/\$C\$3)	= \$C\$6*\$C\$7*G3/\$C\$8	=0.0791/H3^0.25
4	k/D		0			=13	=SQRT(\$C\$5*\$C\$6*\$C\$9/2/\$C\$7/F4/\$C\$3)	= \$C\$6*\$C\$7*G4/\$C\$8	=0.0791/H4^0.25
5	ΔP , psi, Pa		0.25 = B5*144			=14	=SQRT(\$C\$5*\$C\$6*\$C\$9/2/\$C\$7/F5/\$C\$3)	= \$C\$6*\$C\$7*G5/\$C\$8	=0.0791/H5^0.25
6	D, ft		0.5			=15	=SQRT(\$C\$5*\$C\$6*\$C\$9/2/\$C\$7/F6/\$C\$3)	= \$C\$6*\$C\$7*G6/\$C\$8	=0.0791/H6^0.25
7	ρ , lbm/ft^3		62.3			=16	=SQRT(\$C\$5*\$C\$6*\$C\$9/2/\$C\$7/F7/\$C\$3)	= \$C\$6*\$C\$7*G7/\$C\$8	=0.0791/H7^0.25
8	μ , lbm/(ft*s)		0.0006733			=17	=SQRT(\$C\$5*\$C\$6*\$C\$9/2/\$C\$7/F8/\$C\$3)	= \$C\$6*\$C\$7*G8/\$C\$8	=0.0791/H8^0.25
9	g , lbm*ft/(lbf*s^2)		32.174			=18	=SQRT(\$C\$5*\$C\$6*\$C\$9/2/\$C\$7/F9/\$C\$3)	= \$C\$6*\$C\$7*G9/\$C\$8	=0.0791/H9^0.25
10						=19	=SQRT(\$C\$5*\$C\$6*\$C\$9/2/\$C\$7/F10/\$C\$3)	= \$C\$6*\$C\$7*G10/\$C\$8	=0.0791/H10^0.25
11	Volume Flow rate					=10	=SQRT(\$C\$5*\$C\$6*\$C\$9/2/\$C\$7/F11/\$C\$3)	= \$C\$6*\$C\$7*G11/\$C\$8	=0.0791/H11^0.25
12						=11	=SQRT(\$C\$5*\$C\$6*\$C\$9/2/\$C\$7/F12/\$C\$3)	= \$C\$6*\$C\$7*G12/\$C\$8	=0.0791/H12^0.25
13						=12	=SQRT(\$C\$5*\$C\$6*\$C\$9/2/\$C\$7/F13/\$C\$3)	= \$C\$6*\$C\$7*G13/\$C\$8	=0.0791/H13^0.25
14						=13	=SQRT(\$C\$5*\$C\$6*\$C\$9/2/\$C\$7/F14/\$C\$3)	= \$C\$6*\$C\$7*G14/\$C\$8	=0.0791/H14^0.25
15						=14	=SQRT(\$C\$5*\$C\$6*\$C\$9/2/\$C\$7/F15/\$C\$3)	= \$C\$6*\$C\$7*G15/\$C\$8	=0.0791/H15^0.25

$$\dot{V} = A \cdot v = \pi \left(\frac{D}{2} \right)^2 \cdot v$$

$$= 0.154 \text{ ft}^3/\text{s} \times 7.48 \text{ gal/ft}^3 \times 60 \text{ s/min}$$

$$= 69.1 \text{ gal/min}$$

* The difference between this answer and the book probably has to do with how I calculate f .
The book says it read a plot. This could be the difference between the answers.

Method B: Solving via a system of equations

Equation 1: $f = f(Re, \epsilon/D)$

Equation 2: $f = \frac{(Re \cdot F)^2}{Re^2}$

$$f = \frac{0.0791}{Re^{1/4}}$$

$$\sqrt{f} = \frac{\sqrt{0.0791}}{Re^{1/8}}$$

$$Re \cdot F = \frac{D \langle v \rangle \rho}{\mu} \sqrt{\frac{(P_0 - P_1) D}{2 L \rho \langle v \rangle^2}}$$

$$= \frac{D \rho}{\mu} \sqrt{\frac{\Delta P D g}{2 L \rho}}$$

$$= \frac{(0.5 \text{ ft}) \times (62.3 \text{ lbm/ft}^3)}{0.0006733 \frac{\text{lbm}}{\text{ft} \cdot \text{s}}} \sqrt{\frac{(2.5 \text{ lbf}) (0.5 \text{ ft}) (32.174 \frac{\text{lbm} \cdot \text{ft}}{\text{lbf} \cdot \text{s}^2})}{2 (1320 \text{ ft}) (62.3 \frac{\text{lbm}}{\text{ft}^3})}}$$

$$Re \frac{\sqrt{0.0791}}{Re^{1/8}} = 2750$$

$$Re^{7/8} = \frac{2750}{\sqrt{0.0791}} = 97600$$

$$Re = 97600^{8/7} = 36260.45$$

$$F_k = A \cdot f \cdot K$$

A = surface area of the walls of the pipe

$$= \pi D L$$

K = characteristic kinetic energy

$$= \frac{1}{2} \rho \langle v \rangle^2$$

$$F_k = \Delta P A$$

A = cross-sectional area perpendicular to the

pressure force

$$= \pi \left(\frac{D}{2} \right)^2$$

$$\pi D L \cdot f \cdot \frac{1}{2} \rho \langle v \rangle^2 = \Delta P \pi \frac{D^2}{4} \rho$$

$$L \cdot f \cdot \rho \langle v \rangle^2 = \frac{\Delta P D}{2}$$

$$\langle v \rangle^2 = \frac{\Delta P D g}{2 L f \rho}$$

$$\langle v \rangle = \sqrt{\frac{\Delta P D g}{2 L f \rho}}$$

unit check

$$\Delta P = \frac{\text{lbf}}{\text{ft}^2} \quad D = \text{ft}$$

$$\frac{\frac{\text{lbf}}{\text{ft}^2} \cdot \text{ft}}{\frac{\text{lbm}}{\text{ft}^3} \cdot \frac{\text{ft}}{\text{s}^2}} = \frac{\text{ft}^2}{\text{s}^2} \quad \rho = \frac{\text{lbm}}{\text{ft}^3} \quad L = \text{ft} \quad \frac{\text{lbf} \cdot \text{ft}}{\text{lbm} \cdot \text{s}^2} = \frac{\text{ft}^2}{\text{s}^2}$$

$$\langle v \rangle = \frac{Re \cdot \mu}{D \rho} = \frac{(36260.45)(6.73 \times 10^{-4} \frac{\text{lbm}}{\text{ft} \cdot \text{s}})}{(0.5 \text{ ft})(62.3 \frac{\text{lbm}}{\text{ft}^3})}$$

$$= 0.78376 \frac{\text{ft}}{\text{s}}$$

$$V = A \langle v \rangle$$

$$= 0.15389 \frac{\text{ft}^3}{\text{s}} * 7.48 \frac{\text{gal}}{\text{ft}^3} * \frac{60 \text{ sec}}{\text{min}}$$

$$= 69.0663$$

I was unsure of which graph to use. So I used mathematical relationships instead. This is the most likely difference between my answers and the one given in the problem statement.

	A	B	C	D
1	Parameters			
2				
3	L, ft, m		1320	
4	k/D		0	
5	ΔP , psi, Pa	0.25	36	
6	D, ft		0.5	
7	ρ , lbm/ft ³		62.3	
8	μ , lbm/(ft*s)		6.73E-04	
9	g , lbm*ft/(lbf*s^2)		32.174	
10				
11	Volume Flow rate, Method A			
12	V, ft^3/s		0.153891128	
13	V, gal/min		69.06633843	
14				
15				
16				
17	Method B			
18	Resqrt(f)		2745.32	
19	Re^(7/8)		9761.22	
20	Re		36260.45	
21	$\langle v \rangle$, ft/s		0.78376	
22	V, ft^3/s		0.153891128	
23	V, gal/min		69.06633843	
24				
25	Accuracy difference		-8.626E-12	

	A	B	C
1	Parameters		
2			
3	L, ft, m		1320
4	k/D		0
5	ΔP , psi, Pa	0.25	=B5*144
6	D, ft		0.5
7	ρ , lbm/ft ³		62.3
8	μ , lbm/(ft*s)		0.0006733
9	g , lbm*ft/(lbf*s^2)		32.174
10			
11	Volume Flow rate, Method A		
12	V, ft^3/s		=G15*PI()*C6^2/4
13	V, gal/min		=B12*7.48*60
14			
15			
16			
17	Method B		
18	Resqrt(f)		=C6*C7/C8*SQRT(C5*C6*C9/(2*C3*C7))
19	Re^(7/8)		=B18/SQRT(0.0791)
20	Re		=B19*(8/7)
21	$\langle v \rangle$, ft/s		=B20*C8/C6/C7
22	V, ft^3/s		=B21*PI()*C6^2/4
23	V, gal/min		=B22*7.48*60
24			
25	Accuracy difference		=B23-B13

6A.9 Flow of gas through a packed column. A horizontal tube with diameter 4 in. and length 5.5 ft is packed with glass spheres of diameter 1/16 in., and the void fraction is 0.41. Carbon dioxide is to be pumped through the tube at 300K, at which temperature its viscosity is known to be $1.495 \times 10^{-4} \text{ g/cm} \cdot \text{s}$. What will be the mass flow rate through the column when the inlet and outlet pressures are 25 atm and 3 atm, respectively?

Answer: 480 g/s

Ergun Equation:

$$\frac{\Delta P}{L} = \frac{150 \mu (1-\epsilon)^2}{D_p^3 \epsilon^3 \rho} G_0 + \frac{1.75 (1-\epsilon)}{D_p \epsilon^3 \rho} G_0^2$$

$$0 = \frac{1.75 (1-\epsilon)}{D_p \epsilon^3 \rho} G_0^2 + \frac{150 \mu (1-\epsilon)^2}{D_p^3 \epsilon^3 \rho} G_0 - \frac{\Delta P}{L} (100,000)$$

*We will use the arithmetic mean of the density instead of treating it as a function of density. This allows us to integrate the integral

$$G_0 = \frac{\text{kg}}{\text{s}}$$

unit check:

$$\frac{\frac{\text{kg}}{\text{s}}}{\frac{\text{m}^2}{\text{s}^2}} = \frac{\text{kg}}{\text{m}^2 \cdot \text{s}^2}$$

$$\frac{\frac{\text{kg}}{\text{m}^2 \cdot \text{s}}}{\frac{\text{m}}{\text{s}}} = \frac{\text{kg}}{\text{m}^2 \cdot \text{s}^2}$$

$$\frac{\text{bar}}{\text{m}} = \frac{\text{N}}{\text{m}^2}$$

$$\frac{100,000 \frac{\text{N}}{\text{m}^2}}{\text{m}} = \frac{100,000 \text{ N}}{\text{m}^2}$$

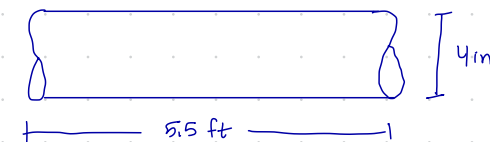
$$\frac{100,000 \text{ N}}{\text{m}^2} = \frac{100,000 \text{ kg}}{\text{m}^2 \cdot \text{s}^2}$$

$$G_0 = 59.056 \frac{\text{kg}}{\text{m}^2 \cdot \text{s}^2} \quad \text{superficial mass flow}$$

$$W = G_0 S_0 = G_0 \pi \left(\frac{D}{2}\right)^2$$

$$= 6.4787 \frac{\text{kg}}{\text{s}}$$

$$= 478.7 \text{ g/s}$$



$$D_p = \frac{1}{16} \text{ in} \quad \epsilon = 0.41 \quad \text{CO}_2 @ 300 \text{ K} \quad \mu = 1.495 \times 10^{-4} \frac{\text{g}}{\text{cm} \cdot \text{s}}$$

$$\Delta P = 25 - 3 \text{ atm} = 22 \text{ atm}$$

$$= 22 \text{ atm} * \frac{1 \text{ bar}}{0.986923 \text{ atm}}$$

$$= 22.292 \text{ bar}$$

$$\rho = \frac{P}{RT} = \frac{PM}{RT} [=] \frac{\text{kg}}{\text{m}^3}$$

$$\rho_{\text{inlet}} = \frac{(22.292 \text{ bar}) \left(44.0 \frac{\text{kg}}{\text{kmol}}\right)}{(0.08314 \frac{\text{bar} \cdot \text{m}^3}{\text{kmol} \cdot \text{K}}) (300 \text{ K})} = 44.64 \frac{\text{kg}}{\text{m}^3}$$

$$\rho_{\text{outlet}} = \frac{(3.0398 \text{ bar}) \left(44.0 \frac{\text{kg}}{\text{kmol}}\right)}{(0.08314 \frac{\text{bar} \cdot \text{m}^3}{\text{kmol} \cdot \text{K}}) (300 \text{ K})} = 5.345 \frac{\text{kg}}{\text{m}^3}$$

$$\bar{\rho} = \rho_{\text{avg}} = \frac{\rho_{\text{inlet}} + \rho_{\text{outlet}}}{2} = 24.942 \frac{\text{kg}}{\text{m}^3}$$

	A	B	C	D	E	F	G
1	Parameters					Coefficient Calculations	
2		Given	SI			a	378.3466663
3	Dtube, m	4	0.1016			b	180.1866422
4	L, m	5.5	1.6764			c	1329724.769
5	ϵ		0.41				
6	μ , kg/(m*s)		0.00001495			Mass flow rate	
7	ρ , inlet, kmol/m^3		1.012264814	44.53965181		G	RHS
8	ρ , outlet, kmol/m^3		0.121471778	5.344758217			
9	P, inlet	25	25.33125685				
10	P, outlet	3	3.039750821				
11	R, bar*m^3/(kmol*K)		0.08341446				
12	Temperature, K		300				
13	Molar Mass, CO2, kg/kmol		44				
14	Average ρ , kg/m^3		24.94220501				
15	Pressure drop		22.29150602				
16	D, pellets	0.0625	0.0015875				

	A	B	C	D	E	F	G
1	Parameters					Coefficient Calculations	
2		Given	SI			a	=1.75*(1-C5)/(C16*C5^3*B14)
3	Dtube, m	4	=B3*0.0254			b	=150*C6*(1-C5)^2/(C16^2*C5^3*B14)
4	L, m	5.5	=B4*0.3048			c	=B15/C4*10^5
5	ϵ		0.41				
6	μ , kg/(m*s)		=0.0001495*100/1000			Mass flow rate	
7	ρ , inlet, kmol/m^3		=C9/B11/B12	=C7*B13		G	RHS
8	ρ , outlet, kmol/m^3		=C10/B11/B12	=C8*B13			
9	P, inlet	25	=B9/0.986923				
10	P, outlet	3	=B10/0.986923				
11	R, bar*m^3/(kmol*K)		0.08341446				=F8*PI()*C3^2/4
12	Temperature, K		300				
13	Molar Mass, CO2, kg/kmol		44				
14	Average ρ , kg/m^3		=(D7*D8)/2				
15	Pressure drop		=C9-C10				
16	D, pellets		=1/16 =B16*0.0254				

6B.2 Friction factor for flow along a flat plate.²

(a) An expression for the drag force on a flat plate, wetted on both sides, is given in Eq. 4.4-30. This equation was derived by using *laminar* boundary layer theory and is known to be in good agreement with experimental data. Define a friction factor and Reynolds number, and obtain the f versus Re relation.

(b) For *turbulent* flow, an approximate boundary layer treatment based on the $1/7$ power velocity distribution gives

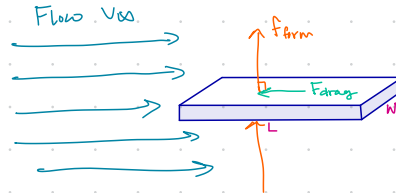
$$F_k = 0.072 \rho v_\infty^2 WL (Lv_\infty \rho / \mu)^{-1/5} \quad (6B.2-1)$$

When 0.072 is replaced by 0.074, this relation describes the drag force within experimental error for $5 \times 10^5 < Lv_\infty \rho / \mu < 2 \times 10^7$. Express the corresponding friction factor as a function of the Reynolds number.

$$= 1.328 \sqrt{\rho \mu L W^2 v_\infty^3} \quad (4.4-30)$$

$$b) \cancel{WL f \rho v_\infty^2} = 0.074 \cancel{\rho v_\infty^2} \cancel{WL} \left(\frac{Lv_\infty \rho}{\mu} \right)^{-1/5}$$

$$f = 0.074 (Re_L)^{-1/5}$$



$$Re_L = \frac{L v_\infty \rho}{\mu}$$

$$F_k = A \cdot f \cdot k \\ = 2(WL) \cdot f \cdot \frac{1}{2} \rho v_\infty^2 \\ = WL f \rho v_\infty^2$$

a)

$$WL f \rho v_\infty^2 = 1.328 \sqrt{\rho \mu L W^2 v_\infty^3}$$

$$f = \frac{1.328 \sqrt{\rho \mu L W^2 v_\infty^3}}{WL \rho v_\infty^2} = 1.328 \sqrt{\frac{\rho \mu L W^2 v_\infty^3}{W^2 L^2 \rho^2 v_\infty^4}}$$

$$= 1.328 \sqrt{\frac{\mu}{L \rho v_\infty}} = \frac{1.328}{\sqrt{Re_L}}$$

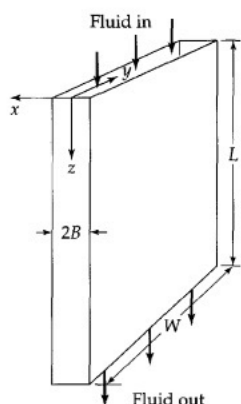
Friction Factor as a function of Reynold's Number Across a Plate:

Laminar flow: $f = \frac{1.328}{\sqrt{Re_L}} \quad 5 \times 10^5 > Re_L$

Turbulent flow: $f = \frac{0.074}{Re_L^{0.12}} \quad 5 \times 10^5 < Re_L < 2 \times 10^7$

6B.3 Friction factor for laminar flow in a slit. Use the results of Problem 2B.3 to show that for the laminar flow in a thin slit of thickness $2B$ the friction factor is $f = 12/Re$, if the Reynolds number is defined as $Re = 2B \langle v_z \rangle \rho / \mu$. Compare this result for f with what one would get from the mean hydraulic radius empiricism.

2B.3 Laminar flow in a narrow slit (see Fig. 2B.3).



$$\begin{aligned} W &= \frac{2}{3} \frac{(\bar{P}_0 - \bar{P}_L) B^3 W \rho}{\mu L} \\ W &= \rho \langle v \rangle A \\ A &= 2BW \\ \langle v \rangle &= \frac{W}{\rho A} = \frac{W}{\rho 2BW} \\ &= \frac{2}{3} \frac{(\bar{P}_0 - \bar{P}_L) B^3 W \rho}{\mu L} \cdot \frac{1}{\rho 2BW} \\ &= \frac{B^2 (\bar{P}_0 - \bar{P}_L)}{3 \mu L} \end{aligned}$$

Fig. 2B.3 Flow through a slit, with $B \ll W \ll L$.

a) $F_k = A \cdot f \cdot k$
 $A = 2(LW)$
 $k = \frac{1}{2} \rho \langle v \rangle^2$

$$F_k = (\bar{P}_0 - \bar{P}_L)(2BW) \quad Re = \frac{2B \langle v \rangle \rho}{\mu}$$

$$2LW \cdot f \cdot \frac{1}{2} \rho \langle v \rangle^2 = 2BW(\bar{P}_0 - \bar{P}_L)$$

$$L \cdot f \cdot \rho \langle v \rangle^2 = 2B(\bar{P}_0 - \bar{P}_L)$$

$$f = \frac{2B(\bar{P}_0 - \bar{P}_L)}{L \cdot \rho \langle v \rangle^2}$$

$$= \frac{2B(\bar{P}_0 - \bar{P}_L)}{L \rho \langle v \rangle^2} \cdot \frac{3 \mu L}{B^3 (\bar{P}_0 - \bar{P}_L)}$$

$$= \frac{6 \mu}{B \rho \langle v \rangle^2}$$

$$= \frac{6}{\frac{Re}{2}} = \frac{12}{Re}$$

b) $R_H = \frac{\text{Netted area}}{\text{Netted perimeter}} = \frac{2BW}{2W + 4B} = \frac{2BW}{2W} \approx B$
 $B \ll W \rightarrow \text{negligible}$

$$f = \frac{16}{Re_h} = \frac{16}{\frac{4B \langle v \rangle \rho}{\mu}} = \frac{4 \mu}{B \langle v \rangle \rho} = \frac{4}{\frac{Re}{2}} = \frac{8}{Re}$$

This shows that this method for laminar flow, is not accurate (mean hydraulic radius)

6B.5 Turbulent flow in horizontal pipes. A fluid is flowing with a mass flow rate $\dot{m}$ in a smooth horizontal pipe of length L and diameter D as the result of a pressure difference $p_0 - p_L$. The flow is known to be turbulent.

The pipe is to be replaced by one of diameter $D/2$ with the same length. The same fluid is to be pumped at the same mass flow rate $\dot{m}$. What pressure difference will be needed?

(a) Use Eq. 6.2-12 as a suitable equation for the friction factor.

(b) How can this problem be solved using Fig. 6.2-2 if Eq. 6.2-12 is not appropriate?

Answer: (a) A pressure difference 27 times greater will be needed.

$$\dot{m} = \dot{m}_1 = \dot{m}_2$$

$$\frac{\rho \langle v_1 \rangle}{\frac{\pi}{4} D^2} = \frac{\rho \langle v_2 \rangle}{\frac{\pi}{16} D^2}$$

$$4 \langle v_1 \rangle = 16 \langle v_2 \rangle$$

$$\langle v_1 \rangle = 4 \langle v_2 \rangle$$

$$f_1 = \frac{0.0729}{Re_1^{0.25}}$$

$$f_2 = \frac{0.0729}{Re_2^{0.25}}$$

$$= \frac{0.0729}{(8 Re_1)^{0.25}}$$

$$= \frac{0.0729}{0.5946 Re_1^{0.25}}$$

$$0.5946 f_2 = \frac{0.0729}{Re_1^{0.25}} = f_1$$

$$Re_1 = \frac{\rho \langle v \rangle D}{\mu}$$

$$Re_2 = \frac{\rho \langle v \rangle \frac{D}{2}}{\mu}$$

$$= \frac{\rho \langle v_1 \rangle D}{8 \mu}$$

$$8 Re_2 = \frac{\rho \langle v_1 \rangle D}{\mu} = Re_1$$

$$A_1 = \frac{\pi}{4} D^2$$

$$A_2 = \frac{\pi}{16} D^2 = \frac{1}{4} A_1$$

$$\begin{aligned} F_L &= A \cdot f \cdot L \\ &= \pi D L \cdot f \cdot \frac{1}{2} \rho v^2 \\ F_L &= (p_0 - p_L) A \\ &= (p_0 - p_L) \frac{\pi}{4} D^2 \\ \pi D L f \cdot \frac{1}{2} \rho v^2 &= (p_0 - p_L) \frac{\pi}{4} D^2 \\ L f \rho v^2 &= (p_0 - p_L) \frac{D}{2} \\ \frac{2 L f \rho v^2}{D} &= p_0 - p_L \end{aligned}$$

Velocity affects:

$$\dot{m} = \rho v A$$

$$= \rho v \frac{\pi}{4} D^2$$

$$v = \frac{4 \dot{m}}{\pi \rho D^2}$$

$$v \propto D^{-2}$$

the mass flow rate is said to be constant. This means when the Diameter shrinks to $D/2$, v increases by 4

Friction Factor Affects:

$$Re = \frac{D v \rho}{\mu}$$

$$Re \propto D \cdot v$$

$$\propto D \cdot D^{-2} = D^{-1}$$

when $D \Rightarrow \frac{D}{2}$ $v \Rightarrow 4v$
Reynolds number double between the two sizes.

$$f = \frac{0.0729}{Re^{0.25}}$$

As diameter increases, Re number dependency becomes D^{-1} .

$$f \propto Re^{-0.25} \propto D^{(-1)(-0.25)}$$

The friction factor depends on $Re^{-1/4}$

$$\Delta P \propto f \cdot \frac{1}{D} \cdot v^2$$

$$(D^{0.25})(D^{-1})(D^{-2})^2$$

$$D^{0.25-1-4} = D^{-4.75}$$

$$= D^{-4.75}$$

Therefore ΔP changes with D by $D^{-4.75}$. Since this change is by $\frac{1}{2}$

$$\Delta P_2 = \left(\frac{1}{2}\right)^{-4.75} \Delta P_1$$

$$= 24.91 \Delta P_1$$

B) 1. Calculate your Reynolds number with D_1 , Re_1

2. The relationship for Reynolds number was shown to be $Re \propto D^{-1}$ so $Re_{new} = 2 Re_{old}$

3. Find the friction factors on the chart: $f_{old} \propto Re_{old}^{-0.25}$ $f_{new} \propto Re_{new}^{-0.25}$

4. Apply the friction factor ratio to find the ratio of the pressure drops:

$$\frac{\Delta P_{new}}{\Delta P_{old}} = \frac{f_{new}}{f_{old}} \cdot \frac{D_{old}}{D_{new}} \cdot \left(\frac{v_{new}}{v_{old}}\right)^2$$

we know this to be 2

we know this to be 4

$$= \frac{f_{new}}{f_{old}} (2)(4)^2$$

$$= 32 \frac{f_{new}}{f_{old}}$$

6B.7 Falling sphere in Newton's drag-law region. A sphere initially at rest at $z = 0$ falls under the influence of gravity. Conditions are such that, after a negligible interval, the sphere falls with a resisting force proportional to the square of the velocity.

(a) Find the distance z that the sphere falls as a function of t .

(b) What is the terminal velocity of the sphere? **Assume that the density of the fluid is much less than the density of the sphere.**

Answer: (a) The distance is $z = (1/c^2g) \ln \cosh cgt$, where $c^2 = \frac{3}{8}(0.44)(\rho/\rho_{\text{sph}})(1/gR)$; (b) $1/c$

The friction factor in Newton's Drag-Law is about $f=0.44$

Also in this region $F_{\text{drag}} \propto v^2$

$$m \frac{dv_z}{dt} = mg - \pi R^2 (0.44) \rho v^2$$

$$\underbrace{\rho_{\text{sphere}} \frac{4}{3} \pi R^3}_{\rho_{\text{sphere}} \frac{4}{3} R} \frac{dv_z}{dt} = \underbrace{\rho_{\text{sphere}} \frac{4}{3} \pi R^3}_{\rho_{\text{sphere}} \frac{4}{3} R} g - \pi R^2 (0.44) \rho v^2$$

$$\frac{dv_z}{dt} = g - \frac{0.44 \rho v^2}{4 \rho_{\text{sphere}} R}$$

$$\text{let } c^2 = \frac{0.44}{4} \frac{\rho}{\rho_{\text{sphere}}} \frac{1}{Rg}$$

$$\frac{dv_z}{dt} = g(1 - c^2 v_z^2)$$

$$\int \frac{dv_z}{1 - c^2 v_z^2} = \int g dt$$

$$\int \frac{dx}{1 - a^2 x^2} = \frac{1}{a} \tanh^{-1}(ax)$$

$$\frac{1}{c} \tanh^{-1}(c v_z) = gt$$

$$t = \frac{1}{cg} \tanh^{-1}(c v_z)$$

$$cgt = \tanh^{-1}(c v_z)$$

$$\tanh(cgt) = c v_z$$

$$v_z(t) = \frac{1}{c} \tanh(cgt)$$

b) Terminal velocity is when $\frac{dv}{dt} = 0$

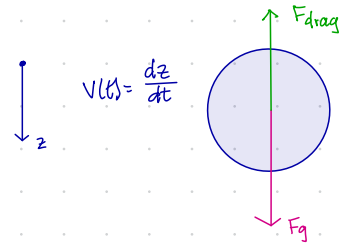
$$\frac{dv_z}{dt} = g(1 - c^2 v_z^2) = 0$$

$$1 - c^2 v_z^2 = 0$$

$$c^2 v_z^2 = 1$$

$$v_z^2 = \frac{1}{c^2}$$

$$v_z = \frac{1}{c}$$



Since we are assuming that $\rho \ll \rho_{\text{sph}}$, buoyancy can be neglected

Newton's 2nd Law (Force Balance):

$$F_z = F_g - F_{kz}$$

$$F_z = m a_z \quad a_z = \frac{dv_z}{dt}$$

$$m_{\text{sph}} = \rho_{\text{sph}} \frac{4}{3} \pi R^3$$

$$F_g = mg$$

$$F_{kz} = A \cdot f \cdot K$$

$$= \pi R^2 (0.44) \left(\frac{1}{2} \rho v^2\right)$$

$$v(t) = \frac{dz}{dt} = \frac{1}{c} \tanh(cgt)$$

$$\int dz = \int \frac{1}{c} \tanh(cgt) dt$$

$$\int \tanh(x) dx = \ln \cosh(x)$$

$$z(t) = \frac{1}{c} \cdot \frac{1}{cg} \ln \cosh(cgt)$$

$$z(t) = \frac{1}{c^2 g} \ln \cosh(cgt)$$